# REGRET IN RUNTIME ANALYSIS OF BLACK-BOX ALGORITHMS

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# Abstract

TODO: Add abstract after completion of report

# Declaration

No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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# Abbreviations & Acronyms

**B | C | E | H | J | M | R | S | T | Y**

**B**

**BBC** Black-Box Complexity. 13

**BBO** Black-Box Optimization. 13, 14, 20, 28, 30, 31, 37

**BO** Bayesian Optimization. 32

**C**

**CI** Continuous Integration. 44

**CLI** Command-Line Interface. 38, 39, 46

**CR** Cumulative Regret. 33, 49

**E**

**EA** Evolutionary Algorithms. 20, 24, 26, 27, 29, 47, 48, 49, 52, 58, 60, 61, 62, 63, 64, 76

**H**

**HIFF** Hierarchical If-and-Only-If. 61

**J**

**JSON** *acr. JavaScript Object Notation.* An open standard file format and data interchange format that uses human-readable text to store and transmit data objects consisting of name–value pairs and arrays (or other serializable values).. 5, 38, 39, 40, 41

**M**

**MAB** Multi-Armed Bandits. 31

**ML** Machine Learning. 13

**R**

**RLS** Randomized Local Search. 25, 42, 56, 58, 59, 60, 61, 63

**S**

**SA** Simulated Annealing. 58

**SR** Simple Regret. 34

**T**

**TDD** Test-Driven Development. 44

**TTFO** Time to First Optimum. 43, 44, 54, 58, 60, 62

**Y**

**YAML** *acr. YAML Ain't Markup Language.* A human-readable data serialization language, commonly used for configuration files.. 38, 39, 40, 41, 46

# Chapter 1

# Introduction

## 1.1 Motivation

Many aspects of our day-to-day lives revolves around finding optimality in problems that we can have no comprehensible insight about. It can be observed in industrial processes of ! TODO!. These types of problems are said to be Black-Box Optimization (BBO) problems.

We do not get much tools to interact with them save from a cost/objective function $\mathbf{f} : \mathcal{X} \mapsto \mathcal{S}$ (the 'oracle') that provides an abstract representation of the problem, where $\mathcal{X}$ is the search space, $\mathcal{S}$ the evaluation space !TODO: find concrete refs!. Typically, $\mathcal{S} \subseteq R$. The task is to minimize or maximize the solutions produced by this objective function, i.e., find a solution that Much study is done in optimizing this cost function without the liberty of classic mathematical tools such as continuity, smoothness and convexity (these methods warrant the oracle to return additional information, such as gradients, apart from the evaluation value of an input [42]).

### 1.1.1 Conventional Solutions

Classical analysis of solving BBO problems are concerned with the so-called Black-Box Complexity (BBC), that is, deriving the runtime bounds for an algorithm to sample a (global) optimum, depending only on the problem size $\mathbf{n}$. The reasoning behind this is palpable; the *computational* complexity of evaluating the objective function for such BBO systems are usually expensive.

An obvious example is Hyperparameter Tuning in Machine Learning (ML)

models !CITE!. The optimal hyperparameters could be thought of as the *optima* for the Black-Box Optimization (BBO) problem of the tuning that maximizes their validation loss[1].

### 1.1.2 The Problem

While this approach secures the bounds on the first sampling of the optimum, it presents no clues on the solution quality produced by the algorithm. Consider the case when the algorithm samples the optimum at time $t$, then in the next step $t+1$, moves completely away from the vicinity. This algorithm might then produce worse answers, asymptotically.

### 1.1.3 Project's Alternative

[1] 'how well the model fits the underlying probability distribution of the data inputs & outputs'

## 1.2 Objectives

### 1.2.1 General Aims

### 1.2.2 Theoretical Exploration

### 1.2.3 Supplementary Software Development

### 1.2.4 Experimental Analysis

## 1.3 Project Outline

### 1.3.1 Literature Study

### 1.3.2 Isolating Relevant Benchmarks

### 1.3.3 Development of Experiment framework

### 1.3.4 Orchestrating Experiments

### 1.3.5 Evaluating Results

### 1.3.6

## 1.4 Ethical Considerations

## 1.5 Report Structure

### 1.5.1 Background

### 1.5.2 Software Design

### 1.5.3 Experiment Structure

### 1.5.4 Evaluation & Discussion of Results

### 1.5.5 Conclusion & Future Work

# Chapter 2

# Background

## 2.1 Computational Problems

### 2.1.1 Problem Formulations

A *computational problem* is a formal specification of a task defined over a set of *problem instances*. Each instance encodes the input data relevant to a particular occurrence of the problem. The *problem class* describes the collection of all such problem instances together with the corresponding output requirements. [28]

Problems are broadly categorized by the nature of their required output.

- A *decision problem* demands a binary answer, whether a solution exists for a problem instance.
- A *search problem* asks for the construction of an explicit solution satisfying a specified property, such as a feasible assignment or a path in a graph, or conclude no solution exists.
- An *optimization problem* further requires that the returned solution be optimal (or near-optimal) with respect to some given objective function.

The decision and search problem variants are closely related because algorithms solving the search variant can always be used to solve the decision variant. [28]

A further distinction concerns the role of randomness in the problem formulation. In a *deterministic* formulation, both the instance and every algorithm operating on it are fully specified without any probabilistic component. In a *stochastic* formulation, the instance itself, the objective evaluations, or the algorithm may

involve randomness. Noisy objective functions, where repeated evaluation of the same candidate solution may return different values, are a canonical example and arise naturally in simulation-based and real-world settings.

### 2.1.2 Algorithms

#### 2.1.2.1 Algorithm Properties

**Deterministic and Randomized Algorithms.**

**Adaptive vs. Non-Adaptive Algorithms**

#### 2.1.2.2 Oracle-Based Computation

An *oracle* is an abstract computational device that answers queries about a problem instance. In optimization, the most common oracle is the *function evaluation oracle*, which, given a query point $x \in \mathcal{X}$, returns the objective value $f(x)$ (or a noisy estimate thereof). The algorithm interacts with the oracle in a *query-response* model: it submits a query and receives a response before deciding on the next query. The total number of such interactions constitutes the *query complexity* of the algorithm, formalized in Section 2.1.5.2 [44, 19].

### 2.1.3 Search Space

#### 2.1.3.1 Structure of the Search Space

The *search space* $\mathcal{X}$ is the set of all candidate solutions considered by an algorithm. Search spaces may be *finite* or *infinite*, and *discrete* or *continuous*.

Discrete combinatorial search spaces, in particular the Boolean hypercube $\{0,1\}^n$, are central to the theoretical analysis of evolutionary algorithms (EA) and other randomized search heuristics [18, 29] used in theoretical BBO analysis. Continuous spaces $\mathbb{R}^d$ arise in numerical optimization [44], and mixed spaces containing both discrete and continuous variables occur in practice but are considerably harder to analyze [5].

The structure of the search space, and in particular how neighboring solutions relate to one another, fundamentally influences the difficulty of optimization. This is encoded by a *neighborhood function* $\mathcal{N} : \mathcal{X} \to 2^{\mathcal{X}}$, assigning to each candidate solution the set of solutions reachable in a single step!CITE: Hoos and Stützle -

2005 - INTRODUCTION!. For the Boolean hypercube, the standard neighborhood consists of all strings at Hamming distance one !CITE: a resource using hamming distance!.

**Remark 1** (Problem domain of interest)**.** *This project is concerned with* Pseudo-Boolean, Combinatorial Optimization problems*, with solution space* $\mathcal{X} \subseteq \{0,1\}^n$*. A candidate solution can be represented as a* bitstring*: a sequence of alphabets* $a \in \{0,1\}$*. The objective functions considered for such problems are of the form* $f : \mathcal{X} \to \mathbb{R}_0^+$*, where* $\mathbb{R}_0^+$ *denotes the non-negative reals. Without loss of generality, an objective function with co-domain* $\mathcal{T} \subseteq \mathbb{R}$ *can be mapped to this convention via the affine shift*

$$g : \mathbb{R} \to \mathbb{R}_0^+, \qquad g(x) = x - \min_{y \in \mathcal{X}} f(y),$$

*which is well-defined whenever* $f$ *attains its minimum on* $\mathcal{X}$*, and preserves the ordering of all objective values by construction.*

#### 2.1.3.2 Constraints and Feasibility

In a *constrained* optimization problem, the *feasible region* $\mathcal{F} \subseteq \mathcal{X}$ is defined by one or more constraints that a valid solution must satisfy [44, 8]. An *unconstrained* problem places no restriction beyond membership in $\mathcal{X}$. Constraints may be *hard* (a solution violating them is inadmissible) or *soft* (violations are penalized in the objective). The boundary of the feasible region can significantly complicate optimization, particularly for local search methods that may generate infeasible intermediate solutions.

### 2.1.4 Objective Functions and Landscape Properties

The geometric structure of the objective function determines both which algorithms are applicable and what convergence guarantees are attainable. A central organizing principle is that access to higher-order analytical information about $f$, i.e., gradients, curvature, or structural properties such as convexity, generally enables stronger theoretical guarantees and more efficient algorithms; restricting this access, as in the black-box setting, correspondingly limits what can be guaranteed.

#### 2.1.4.1 Zeroth-, First-, and Second-Order Methods

Optimization algorithms can be classified by the order of analytical information they require from $f$. This *information hierarchy* has three tiers.

*Zeroth-order* methods use only evaluations of $f(x)$, with no gradient or curvature information. The algorithm observes only the output of the function evaluation oracle at each queried point and must infer search directions from comparisons of objective values alone. This exactly describes the black-box model formalized in Section 2.2: when the internal structure of $f$ is inaccessible or analytically intractable, zeroth-order methods are the only option. Their weaker information access is the fundamental reason black-box optimization is harder in general than white-box optimization, and it is the regime in which all algorithms studied in this project operate.

*First-order* methods additionally access the gradient $\nabla f(x) \in \mathbb{R}^d$, which encodes the direction of steepest ascent at $x$. For a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$, gradient-based methods exploit this to define descent directions and form the backbone of classical continuous optimization [44]. When $f$ is convex but non-differentiable, the gradient generalizes to the *subdifferential* $\partial f(x)$, the set of all *subgradients* $g$ satisfying

$$f(y) \geq f(x) + g^\top (y - x) \quad \forall y \in \mathbb{R}^d.$$

Subgradient methods replace the gradient with any element of $\partial f(x)$, recovering convergence under convexity alone, albeit at rates slower than those attainable with smooth gradients [8].

*Second-order* methods further access the Hessian $\nabla^2 f(x) \in \mathbb{R}^{d \times d}$, which captures local curvature. Methods that utilize or approximate the Hessian, such as Newton and quasi-Newton schemes, can achieve superlinear or quadratic convergence near a solution [44], at the cost of maintaining a $d \times d$ matrix.

Together, these three tiers define distinct levels of oracle access, and their availability directly governs the choice and efficiency of applicable algorithms. The move from second-order to zeroth-order corresponds to progressively weaker assumptions about what the oracle reveals, culminating in the black-box model where only function values are observable.

#### 2.1.4.2 Landscape Properties

The following geometric properties of $f$ characterize which tier of analytical information is both available and useful.

**Convexity.** A function $f : \mathbb{R}^d \to \mathbb{R}$ is *convex* if for all $x, y \in \mathbb{R}^d$ and $\lambda \in [0,1]$ [8],

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y).$$

Convexity guarantees that every local minimum is global, rendering the landscape tractable for first-order methods. *Strong convexity* additionally requires

$$f(y) \geq f(x) + \nabla f(x)^\top (y-x) + \tfrac{\mu}{2}\|y-x\|^2$$

for some $\mu > 0$, implying a unique global minimum and linear convergence rates for gradient descent !CITE: Nesterov 2004!.

**Smoothness and Lipschitz Continuity.**

**Ruggedness and Multimodality.**

### 2.1.5 Complexity Measures

#### 2.1.5.1 Classical Complexity: Time and Space

#### 2.1.5.2 Information Complexity: Query and Oracle Complexity

#### 2.1.5.3 Expected vs. Worst-Case Analysis

## 2.2 Black-Box Optimization

### 2.2.1 The Black-Box Model

In the *black-box optimization* model, the algorithm has no access to the analytical form of the objective function. It interacts with the objective exclusively through an oracle that, upon receiving a query $x \in \mathcal{X}$, returns $f(x)$ (or a noisy estimate thereof) [19, 16, 41]. This constitutes *zeroth-order* information: no gradient or higher-order information is available. The black-box model is appropriate whenever the objective arises from a simulation, physical experiment, or any process whose internal structure is inaccessible or too complex to exploit analytically.

In contrast, in *white-box optimization*, information about the problem instance is make available to the algorithm. This enables algorithms usage of some computational tools such as gradients, supporting optimization convergence.

#### 2.2.1.1 Variants of Black-Box Models

### 2.2.2 Black-Box Complexity

### 2.2.3 Purpose of Randomized Search Heuristics

#### 2.2.3.1 No-Free-Lunch Theorems

#### 2.2.3.2 Taxonomy of Derivative-Free Methods

### 2.2.4 Randomized Search Heuristic Algorithms

#### 2.2.4.1 Evolutionary Algorithms (EA)

Evolutionary algorithms are a broad family of population-based, randomized search heuristics inspired by biological evolution. In discrete optimization, the canonical paradigm maintains one or more *candidate solutions* (bit strings) and iterates two operators: *mutation*, which perturbs a solution stochastically, and *selection*, which retains solutions according to their objective value [18, 16]. The simplest instance of this paradigm, the $(1+1)$ EA, serves as the theoretical baseline throughout this project.

**The $(1+1)$ EA.** The $(1+1)$ EA maintains a single current solution $x \in \{0,1\}^n$. In each iteration it produces one offspring $y$ by *standard bit mutation*: every bit

is flipped independently with probability $p$. The offspring replaces the parent if and only if $f(y) \geq f(x)$ (*elitist selection*). With the standard choice $p = 1/n$ the expected number of flipped bits is exactly one. The resulting process is a homogeneous Markov chain on $\{0,1\}^n$ with a non-decreasing fitness sequence [18, 54].

**Algorithm 1** The $(1+1)$ EA with mutation probability $p \in (0,1)$

```
Sample x ∈ {0,1}^n uniformly at random; evaluate f(x)
repeat
    y ← x
    for i = 1,...,n do
        flip y_i with probability p, independently
    end for
    if f(y) ≥ f(x) then x ← y
    end if
until termination criterion met
```

The classical runtime results for $p = 1/n$ are

$$\mathbb{E}[T_{\textsc{OneMax}}] = e\,n \ln n + O(n) \qquad \text{and} \qquad \mathbb{E}[T_{\textsc{LeadingOnes}}] = \tfrac{e-1}{2}\,n^2 + o(n^2),$$

and the $\Theta(n \log n)$ bound extends to *all* linear pseudo-Boolean functions with non-zero weights [54]. On general landscapes the algorithm may require exponential time: even simple quadratic and unimodal functions can trap it indefinitely without restarts or adaptive mutation [18]. Advanced variants and implementation-aware refinements are discussed in Appendix A.

#### 2.2.4.2 Simulated Annealing

**Cooling Schedules**

#### 2.2.4.3 Randomized Local Search (RLS)

**Stochastic Hill-Climber - Greedy RLS**

$\epsilon - greedy$ **RLS**

**Decaying** $\epsilon - greedy$ **RLS**

### 2.2.5 Few Pseudo-Boolean Problems

The *pseudo-Boolean* setting considers functions $f : \{0,1\}^n \to \mathbb{R}$ defined on binary strings of length $n$. Several benchmark functions are widely used to study and compare the behavior of randomized search heuristics, each encoding a different type of difficulty. [41, 19]

#### 2.2.5.1 OneMax

The OneMax function counts the number of one-bits in a bit string:

$$\textsc{OneMax}(x) = \sum_{i=1}^{n} x_i, \qquad x \in \{0,1\}^n.$$

The unique global maximum is $x^* = 1^n$ with $\textsc{OneMax}(1^n) = n$. OneMax is the canonical baseline benchmark: the landscape is unimodal and every improving move increases the function value by exactly one, providing a clear gradient for local search methods. The expected runtime of the $(1+1)$ Evolutionary Algorithms (EA) on OneMax is $\Theta(n \log n)$. !CITE: Droste, Jansen, Wegener (2002)!

#### 2.2.5.2 LeadingOnes

The LeadingOnes function counts the length of the maximal leading prefix of ones:

$$\textsc{LeadingOnes}(x) = \sum_{i=1}^{n} \prod_{j=1}^{i} x_j.$$

Equivalently, $\textsc{LeadingOnes}(x)$ equals the index of the first zero bit minus one, or $n$ if no zero exists. The unique global optimum is again $1^n$. Because progress requires the leading bit to be set before any subsequent bit can contribute, LeadingOnes is significantly harder than OneMax: the expected runtime of the $(1+1)$ EA is $\Theta(n^2)$. !CITE: Droste, Jansen, Wegener (2002)!

#### 2.2.5.3 Jump

The Jump function with parameter $k \in \{1, \ldots, n\}$ coincides with OneMax on bit strings with at most $n-k$ ones and on the global optimum $1^n$, but introduces a region of reduced fitness for strings with between $n-k+1$ and $n-1$ ones. This creates a "gap" of width $k$ that algorithms relying on single-bit improvements cannot cross: the set of strings with exactly $n-k$ ones constitutes a local optimum

of standard local search, and reaching the global optimum $1^n$ requires flipping all $k$ remaining zero bits simultaneously. For the $(1+1)$ EA with standard bit mutation at rate $1/n$, the probability of a specific $k$-bit flip is $\Theta(n^{-k})$, giving an expected runtime of $\Theta(n^k)$ on $\textsc{Jump}_k$.

Formally, $\textsc{Jump}_{k,n}\colon \{0,1\}^n \to \mathbb{R}$ is defined as

$$\textsc{Jump}_{k,n}(x) = \begin{cases} k + \sum_{i=1}^{n} x_i & \text{if } \sum_{i=1}^{n} x_i \leq n-k \text{ or } \sum_{i=1}^{n} x_i = n, \\ n - \sum_{i=1}^{n} x_i & \text{otherwise.} \end{cases}$$

Droste, Jansen, and Wegener [18] proved that the expected running time of the $(1+1)$ EA on $\textsc{Jump}_{k,n}$ is $\Theta(n^k + n\log n)$ for $k \in \{1,2,\ldots,n\}$, which simplifies to $\Theta(n^k)$ for $k \geq 2$.

#### 2.2.5.4 BinaryValue[BinVal]

The BinaryValue (BinVal) function interprets a bit string as a binary integer by weighting bits exponentially by position:

$$\textsc{BinVal}(x) = \sum_{i=1}^{n} 2^{i-1} x_i.$$

The global optimum is $1^n$ with value $2^n - 1$. The exponential weighting means that the most significant bit contributes more than the combined value of all less significant bits, making BinVal highly structured.

#### 2.2.5.5 Trap

#### 2.2.5.6 Plateau

#### 2.2.5.7 Hierarchical If-Only-If

#### 2.2.5.8 MaxSAT

*Maximum Satisfiability* (MaxSAT) is the optimization variant of the Boolean satisfiability problem (SAT). Given a propositional formula in conjunctive normal form (CNF) !TODO: Cite or explain CNF! with $m$ clauses over $n$ Boolean variables, the goal is to find a truth assignment that maximizes the number of satisfied clauses. MaxSAT is NP-hard in general [30] and serves as an important

benchmark for combinatorial optimization algorithms.

#### 2.2.5.9 Tunable Landscapes: NK-Landscapes

*NK-landscapes* [37] are a parametric family of pseudo-Boolean functions controlled by two integers: $N$ (the number of variables) and $K$ (the number of epistatic interactions per variable, $0 \leq K \leq N-1$). The fitness of a string $x \in \{0,1\}^N$ is defined as

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i\Big(x_i, x_{\pi(i,1)}, \ldots, x_{\pi(i,K)}\Big),$$

where each local contribution $f_i : \{0,1\}^{K+1} \to [0,1]$ is drawn independently and uniformly at random, and $\pi(i,\cdot)$ denotes the $K$ epistatic neighbours of variable $i$.

When $K = 0$ the contributions are independent, producing a smooth, additively separable landscape analogous to OneMax. As $K$ increases toward $N-1$, epistatic interactions intensify and the landscape becomes increasingly rugged, with the number of local optima growing in the limit. NK-landscapes thus provide a tunable model of ruggedness and are widely used to study algorithmic scalability and robustness across a spectrum of problem difficulties.

### 2.2.6 Traditional Performance Measures in BBO

The theoretical analysis of BBO algorithms has developed several complementary frameworks for quantifying performance. Each framework fixes one dimension of the problem, either the evaluation budget or the target solution quality, and asks what can be guaranteed along the other. Together they form the standard vocabulary for comparing randomized search heuristics, yet all share a common limitation that motivates the regret-based view developed in Section 2.4.

#### 2.2.6.1 Expected Runtime (Hitting Time)

The predominant performance measure in the runtime analysis of evolutionary algorithms is the *expected optimization time*: the expected number of function evaluations until an optimal solution is queried for the first time [18, 16]. Formally, for a black-box algorithm producing query sequence $x_1, x_2, \ldots$, the *first hitting*

*time* of the global optimum is

$$\tau_{f^*} := \min\Big\{t \in \mathbb{N} : f(x_t) = f^*\Big\},$$

and the expected runtime is $\mathbb{E}[\tau_{f^*}]$. This quantity has proved immensely useful for establishing complexity hierarchies and ranking algorithms; the results $\Theta(n \log n)$ for the $(1+1)$ EA on OneMax, $\Theta(n^2)$ on LeadingOnes, and $\Theta(n^k)$ on $\text{Jump}_k$ are canonical examples [18]. The primary analytical tools for deriving such bounds are drift theory [41], the fitness-level method, and Markov chain arguments [16].

#### 2.2.6.2 Fixed-Budget Analysis

In *fixed-budget analysis* [35], the evaluation budget $T$ is fixed in advance and the quantity of interest is the expected quality of the best solution found within that budget. Denoting by $f_T^*(\mathcal{A})$ the best objective value achieved by algorithm $\mathcal{A}$ within $T$ evaluations, the fixed-budget framework characterizes $\mathbb{E}[f_T^*(\mathcal{A})]$ as a function of $T$ [35]. This perspective is practically motivated: in almost all realistic settings a user cannot detect when the global optimum has been found, and the algorithm must simply return the best solution found when the budget is exhausted [47]. As a consequence, comparing two algorithms under a fixed budget may reverse the conclusions drawn from comparing their expected runtimes: an algorithm with a larger $\mathbb{E}[\tau_{f^*}]$ may still produce superior intermediate solutions within a given budget [35]. This framework connects directly to the simple regret defined in Section 2.4.2.3: maximizing $\mathbb{E}[f_T^*(\mathcal{A})]$ is equivalent to minimizing expected simple regret at budget $T$.

#### 2.2.6.3 Fixed-Target Analysis and Runtime Profiles

*Fixed-target analysis* generalizes the expected runtime by considering not only the global optimum but every intermediate fitness level. For a target quality $v$, the *first hitting time* is

$$\tau_v := \min\Big\{t \in \mathbb{N} : f(x_t) \geq v\Big\},$$

and the goal is to characterize the distribution of $\tau_v$ for all relevant $v$. This viewpoint appears in the analysis of restricted black-box models; for example,

Doerr and Lengler [17] derive hitting-time bounds across all fitness levels for OneMax under several information-theoretic restrictions, revealing how model constraints shift the difficulty at each tier of the landscape.

Pinto and Doerr [47] formalize this into the *runtime profile* framework: the *expectation-based runtime profile* $\left(\mathbb{E}[\tau_v]\right)_v$ and the strictly finer *distributional runtime profile* $\mathcal{P}(v,t) := \mathbb{P}(\tau_v > t)$ together capture the full anytime behavior of an algorithm rather than only its behavior at the endpoint. Crucially, runtime profiles can identify *crossover fitness levels* where one algorithm overtakes another in the quality trajectory, behavior invisible to the single expected optimization time. The formal development, including the connection between distributional runtime profiles and cumulative regret, is given in Section 4.3.

#### 2.2.6.4 Limitations and the Case for a Continuous Performance Measure

Despite their individual strengths, all three frameworks above share a structural limitation: they measure algorithm performance at isolated thresholds rather than over the entire quality trajectory. The expected runtime reduces the run to a single hitting event; fixed-budget analysis reduces it to a single scalar per budget value; and even the runtime profile, while richer, is a collection of per-level summaries rather than a unified measure of trajectory quality. Crucially, none of these measures penalizes an algorithm for spending many evaluations at low fitness values: a strategy that stagnates near the starting fitness and then discovers the optimum in a single lucky step is indistinguishable from one that makes steady, monotone progress, as far as $\mathbb{E}[\tau_{f^*}]$ is concerned.

Pinto and Doerr [47] identify this gap explicitly, calling for *combined measures* capable of describing the anytime behavior of an evolutionary algorithm within a unified framework, and naming regret-based measures from machine learning as a natural candidate, while noting that a fully satisfying formulation remains an open problem. This project takes that call as its starting point. The sequential decision-making framework and the formal vocabulary of regret are developed in Sections 2.3–2.4; their synthesis with BBO performance analysis is the subject of Section 2.5.

## 2.3 Sequential and Online Optimization Frameworks

### 2.3.1 Exploration-Exploitation Trade-off

Sequential black-box optimization is governed by the exploration–exploitation trade-off. At each round, the algorithm must either explore uncertain regions of the search space to gain information or exploit the currently best-known region to obtain good values immediately [13, 10, 40] Since each evaluation is both a measurement and a search step, neither extreme is satisfactory: pure exploitation can lock onto a suboptimal region, while pure exploration wastes budget.

This tension is central to regret-based analysis. Regret measures the cost of not knowing the objective in advance, so it captures both slow identification of good regions and repeated evaluation of poor ones.

### 2.3.2 Multi-Armed Bandits

The MAB model is the basic framework for sequential decision-making under partial feedback [10, 40]. In the stochastic $K$-armed setting, an algorithm chooses an arm $A_t \in \{1, \ldots, K\}$ at each round and observes only the reward of that arm. If arm $i$ has mean reward $\mu_i$ and $\mu^* = \max_i \mu_i$, the standard pseudo-regret is

$$\bar{R}_T = T\mu^* - \sum_{t=1}^{T} \mu_{A_t} = \sum_{i:\Delta_i>0} \Delta_i \, \mathbb{E}[N_i(T)],$$

where $\Delta_i = \mu^* - \mu_i$ is the suboptimality gap and $N_i(T)$ is the number of pulls of arm $i$ up to time $T$ [10, 40]. Regret is therefore driven by how often the learner samples suboptimal actions.

This model connects naturally to BBO. In a finite search space, candidate solutions can be viewed as arms and objective evaluations as arm pulls. For large or structured domains, however, this flat view breaks down, motivating more structured models such as Bayesian optimization.

### 2.3.3 Reinforcement Learning (Overview)

### 2.3.4 Bayesian Optimization

Bayesian Optimization (BO) treats expensive black-box optimization as a sequential learning problem. The unknown objective is modelled by a probabilistic surrogate, typically a Gaussian process, and an acquisition function selects the next query by balancing predicted value against uncertainty [36, 50, 21].

For this chapter, Bayesian optimization matters mainly because it is the clearest bridge between regret theory and black-box optimization. In particular, GP-UCB admits sublinear cumulative-regret guarantees under Gaussian-process or RKHS[1] assumptions [52, 14, 49]. At the same time, these guarantees rely on surrogate assumptions that are much more natural in low-dimensional continuous spaces than in pseudo-Boolean domains such as $\{0,1\}^n$. This makes regret an established concept in black-box optimization, but not yet one that transfers directly to the combinatorial heuristics studied here.

[1] *abbr.* Reproducing kernel Hilbert space. (Optionally see [22])

## 2.4 Computational Regret

### 2.4.1 Definition of Regret

Let $f : \mathcal{X} \to \mathbb{R}$ be an objective function with global optimum value $f^* = \max_{x \in \mathcal{X}} f(x)$. An algorithm that queries points $x_1, x_2, \ldots, x_T$ incurs *regret* by querying suboptimal solutions. Regret quantifies the gap between the algorithm's performance and that of an oracle always querying the optimum [13]. !TODO: add more context with reference to MAB & rewards!

### 2.4.2 Types of Regret

#### 2.4.2.1 Instantaneous Regret

The *instantaneous regret* at round $t$ is

$$\rho_t = f^* - f(x_t),$$

capturing the suboptimality of the single query made at step $t$. A related variant tracks the *incumbent* (best-so-far) solution rather than the current query:

$$\rho_t^{\text{inc}} = f^* - f(\hat{x}_t), \quad \hat{x}_t = \arg\max_{s \le t} f(x_s).$$

Because $\hat{x}_t$ is non-decreasing in quality, $\rho_t^{\text{inc}}$ is monotonically non-increasing over $t$, and is equivalent to simple regret evaluated at time $t$( 2.4.2.3).

#### 2.4.2.2 Cumulative Regret

Two cumulative regret variants arise naturally and serve different purposes.

The *Cumulative Regret (CR)* after $T$ rounds is [13]

$$R_T = \sum_{t=1}^{T} \rho_t = \sum_{t=1}^{T} \Big( f^* - f(x_t) \Big).$$

This is the standard formulation from the online learning literature: it sums the instantaneous suboptimality of each individual query, regardless of what the algorithm has previously observed. Minimizing $R_T$ corresponds to performing well *throughout* the optimization process, penalizing every suboptimal query equally.

The *incumbent cumulative regret* instead tracks the best-so-far process

$$B(t) := \max_{s \le t} f(x_s),$$

yielding

$$R_T^{\text{inc}} = \sum_{t=1}^{T} \rho_t^{\text{inc}} = \sum_{t=1}^{T} \Big( f^* - B(t) \Big).$$

Since $B(t)$ is non-decreasing in $t$, the per-step contributions $\rho_t^{\text{inc}}$ are non-increasing, and $R_T^{\text{inc}} \le R_T$ for all $T$. The incumbent formulation is the appropriate notion when the practitioner retains the best solution found rather than only the most recent query, which is the standard convention for the randomized search heuristics studied in this project. All regret analyses in Chapter 5 use $R_T^{\text{inc}}$ unless otherwise stated.

In continuous time, the incumbent cumulative regret is approximated via a left-hold (piecewise-constant) integral

$$R_T^{\text{inc}} \approx \sum_{t=1}^{T} \rho_{t-1}^{\text{inc}} \, \Delta t_t, \quad \Delta t_t = t - t_{t-1},$$

which accounts for unequal spacing between evaluations when function evaluation time is non-uniform. In all experiments reported here, evaluations are unit-cost and time is measured in discrete query counts, so $\Delta t_t = 1$ throughout and this reduces to the standard discrete sum.

#### 2.4.2.3 Simple Regret

The *Simple Regret (SR)* after $T$ rounds is [11]

$$r_T = f^* - f(\hat{x}_T),$$

where $\hat{x}_T = \arg\max_{t \le T} f(x_t)$ is the best solution encountered. Simple regret measures the quality of the *final output* and is the appropriate notion when the goal is to return a good solution rather than to perform well during the search. Note that $r_T = \rho_T^{\text{inc}}$: simple regret is the incumbent instantaneous regret at the final round.

When comparing algorithms across multiple independent runs, the standard

scalar summary is the *expected simple regret*

$$\mathbb{E}[r_T] = \frac{1}{N}\sum_{i=1}^{N} r_T^{(i)},$$

where $r_T^{(i)}$ denotes the simple regret of the $i$-th run.

#### 2.4.2.4 Normalized Regret

To enable comparison across problem instances with different objective scales, simple regret can be normalized by the range of attainable fitness values:

$$\tilde{r}_T = \frac{f^* - f(\hat{x}_T)}{f^* - f_{\text{worst}}} \in [0, 1],$$

where $f_{\text{worst}}$ is the worst observed or theoretical fitness value. A value of 0 indicates an optimal solution was found; a value of 1 indicates no improvement over the worst case. When $f^* = f_{\text{worst}}$ (a degenerate constant landscape), the problem is trivially solved and normalized regret is defined as 0.

### 2.4.3 Regret Bounds

Regret analysis studies how regret grows with the budget $T$. The central benchmark is sublinear regret: if $R_T = o(T)$, then the average regret $R_T/T$ tends to zero, meaning that the algorithm asymptotically performs as well as an oracle with full information [13, 10, 40].

In finite stochastic bandits, classic upper bounds show that algorithms such as UCB1 achieve logarithmic gap-dependent regret and sublinear worst-case regret [6]. Matching lower-bound results show that logarithmic regret is, in general, unavoidable up to constants under standard assumptions [39]. In Gaussian-process bandits, both upper and lower bounds depend on information gain and kernel structure rather than only on the number of actions [52, 14, 49]. These results are important here mainly as a template: regret can be treated as a complexity measure for sequential optimization.

### 2.4.4 Connections to Runtime Analysis

## 2.5 Regret-Aware Black-Box Optimization

Regret provides a fine-grained complexity measure in the black-box setting: rather than simply asking whether an algorithm finds the optimum within $T$ queries, regret tracks how suboptimal the queries are throughout the process. This is particularly relevant when every evaluation is costly and performing well before the budget is exhausted matters in practice.

### 2.5.1 Regret as a Performance Measure for Randomized Search Heuristics

For randomized search heuristics, regret complements rather than replaces runtime analysis. Runtime asks when a target is first reached; regret asks how good the algorithm's queried or retained solutions are throughout the run. Fixed-budget analysis already moves in this direction, since expected simple regret is just the expected final suboptimality after a budget of $T$ evaluations [35, 11].

The distinction matters operationally. Low simple regret is appropriate when only the final returned solution matters. Low cumulative regret is appropriate in anytime settings, where useful intermediate performance also matters [13, 47]. For the heuristics considered in this project, the incumbent-based formulation is the natural one because the algorithm's output at interruption is the best-so-far solution.

### 2.5.2 Related Work: Regret in Bayesian and Surrogate-Based Optimization

### 2.5.3 Related Work: Regret-Inspired Analysis of Evolutionary Algorithms

### 2.5.4 Related Work: Bandit-Inspired Metaheuristics

### 2.5.5 Connections to Runtime Analysis via Drift Theory

### 2.5.6 Open Questions and Project Scope

# Chapter 3

# Supplementary Software Design

This chapter discusses the supplementary software developed for carrying out multiple experiments on *simulated* BBO problems efficiently. Trivially, analysis cannot be carried out on truly black-box instances; the problems are emulated to be black-box by providing only the *zeroth-order* information.

## 3.1 Purpose of the Software

A supplementary experimental framework, `regret`, was developed to support the orchestration and analysis of experiments on pseudo-Boolean and combinatorial optimization problems. The framework was motivated by the absence of readily available tooling that combines (i) flexible, declarative batch experiment execution, (ii) trajectory-level data collection, and (iii) regret-specific analysis pipelines in a single cohesive system.

Existing platforms such as Google Vizier [23] and EvoX [31] address hyperparameter optimization and population-based search respectively, but neither provides native support for regret metrics, runtime profile analysis, or the configuration-first experiment execution model required for this project's experimental protocol. `regret` was therefore purpose-built to fill this gap, providing a reproducible, extensible environment for studying simple, instantaneous, and cumulative regret across a controlled suite of algorithms and benchmark problems.

### 3.1.1 Preexisting software

Several established frameworks were considered prior to development. **Google Vizier** [23] provides a scalable black-box optimization service with a client–server architecture, but is oriented toward hyperparameter tuning rather than controlled algorithmic benchmarking, and does not expose trajectory-level data in a form amenable to regret analysis. **EvoX** [23] is a JAX-based evolutionary computation library offering GPU-accelerated population methods, but its abstraction layer couples algorithm state to framework internals in ways that preclude the lightweight callback-driven trajectory recording required here. General-purpose scientific workflow systems such as Snakemake [43] and Luigi [20] offer pipeline orchestration but impose significant configuration overhead for the combinatorial (algorithm × problem × budget) sweeps central to this study.

### 3.1.2 Capabilities

The framework supports the full experimental life-cycle through a single Command-Line Interface (CLI), `run_experiment`, which accepts one or more YAML configuration files and executes the appropriate sub-command. Table 3.1 summarizes the available commands.

Table 3.1: CLI sub-commands exposed by the `run_experiment` entry point (equivalently `python -m regret`).

| Command | Description |
|---|---|
| `validate` | Parse and validate one or more YAML configurations against the JSON Schema and semantic registry checks without executing any computation. |
| `plan` | Print a dry-run summary: problem/algorithm combinations, total runs, and evaluation budgets. |
| `run [--no-plot]` | Execute one or more experiment configurations; optionally suppress plot generation after completion. |
| `analyze` | Regenerate all plots from previously saved JSON results without re-running experiments. |
| `table [--format]` | Generate summary tables from existing results in `latex`, `csv`, or `markdown` format. |

**Batch Experiment Execution.** All experiments are specified declaratively via YAML configuration files validated against a JSON Schema prior to execution.

A configuration defines the suite name, evaluation budgets, run count, output mode, and lists of problem and algorithm specifications. Problems and algorithms are referenced by string keys resolved against internal registries, providing a clean separation between configuration and implementation.

**Parallelised execution.** When `parallel: true` is set in the YAML configuration, independent runs are initiated via Python's `ProcessPoolExecutor`, with per-run random seeds fixed to indices $0, 1, \ldots, r-1$ to ensure full reproducibility. Results are reordered by seed index after completion.

**Flexible output modes.** In `lite` mode, only summary statistics are persisted per run. In `full` mode, the complete evaluation trajectory (a sequence of `(evaluation_index, current_value, best_value)` tuples) is recorded and stored alongside aggregate statistics in structured JSON files.

**Batched multi-configuration execution.** All sub-commands accept multiple configuration files, enabling sequential execution of logically grouped experiment suites from a single invocation.

### 3.1.3 Design Decisions

The software was intended to be extensive & configurable for other developers and users to interact with quickly, with options for modularity when required. This was enabled by the adoption of configuration-first, job dispatch policy for independent and declarative experimentation. Problems and Algorithms to test against can easily be defined by extending abstract classes `Problem` & `Algorithm`, ensuring a common interface for consistent interactions.

#### 3.1.3.1 Independent Job Execution framework

The CLI is implemented using **Typer** [48], which maps sub-commands to typed Python functions with automatic argument parsing. The "orchestration" layer coordinates problem & algorithm instantiation, result persistence, and plot generation, delegating individual runs to `ExperimentRunner`. Parallel execution uses `ProcessPoolExecutor` from Python's standard `concurrent.futures` module; results are collected via `as_completed` and reordered by seed for deterministic

output ordering. Figure 3.1 illustrates the full validation and the experiment (job) dispatch pipeline.

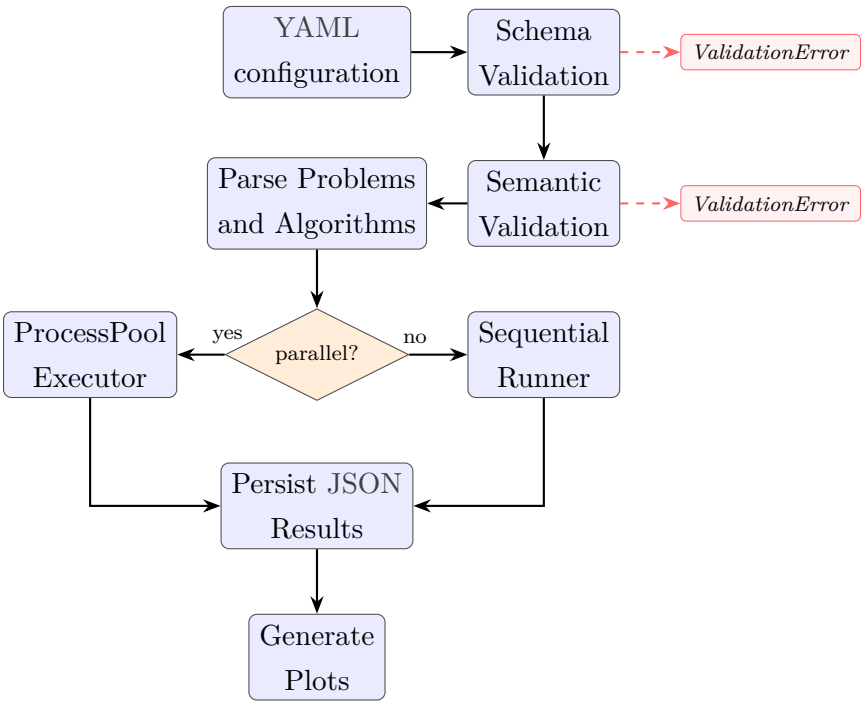



Figure 3.1: Validation and job dispatch pipeline. YAML configuration files are validated structurally via JSON Schema and then semantically against class registries before execution begins. Any validation failure raises a `ValidationError` and halts the pipeline before computation begins.

#### 3.1.3.2 Raw Output JSON files

The output JSON files are *compacted* by replacing redundant stylistic whitespaces and newlines with other seperators. This reduced the raw file sizes by upto 50%. JSON files were preferred for storing the raw files since it included *mixed* data, including metadata, general statistics for each run, along with the trajectories. A 'NoSQL' (key-value) database was considered but not used since the experiments were relatively small scale.

#### 3.1.3.3 UML Diagram

Figure 3.2 presents the core class structure of the framework. Abstract base classes `Problem` and `Algorithm` define the shared interface enforced across all concrete implementations. Concrete subclasses are registered by string key and

resolved at runtime. `ExperimentRunner` orchestrates individual runs and delegates result persistence, while the `CoolingSchedule` protocol is resolved from YAML at parse time via a *registry*.

### 3.1.4 Limitations

The framework is currently constrained to **binary search spaces** $\{0,1\}^n$; continuous and integer-valued domains are not supported. **Multi-objective optimization** is not implemented. The NK landscape optimum computation uses exhaustive search, restricting its practical use to $n \leq 20$. Runtime profile analysis assumes integer-valued, unit-increment fitness functions for the exact tail-sum identity:

$$\mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \sum_{t'=1}^{T} \Big[1 - P(\tau_v \leq t')\Big];$$

continuous fitness functions (e.g.`NK-Landscape`) yield approximate results and may exhibit minor discrepancies in the $\mathbb{E}[\mathrm{CR}(T)]$ verification plot.

### 3.1.5 Result Interpretations

All experiment output is written under a structured directory hierarchy. Raw JSON result files are stored at:

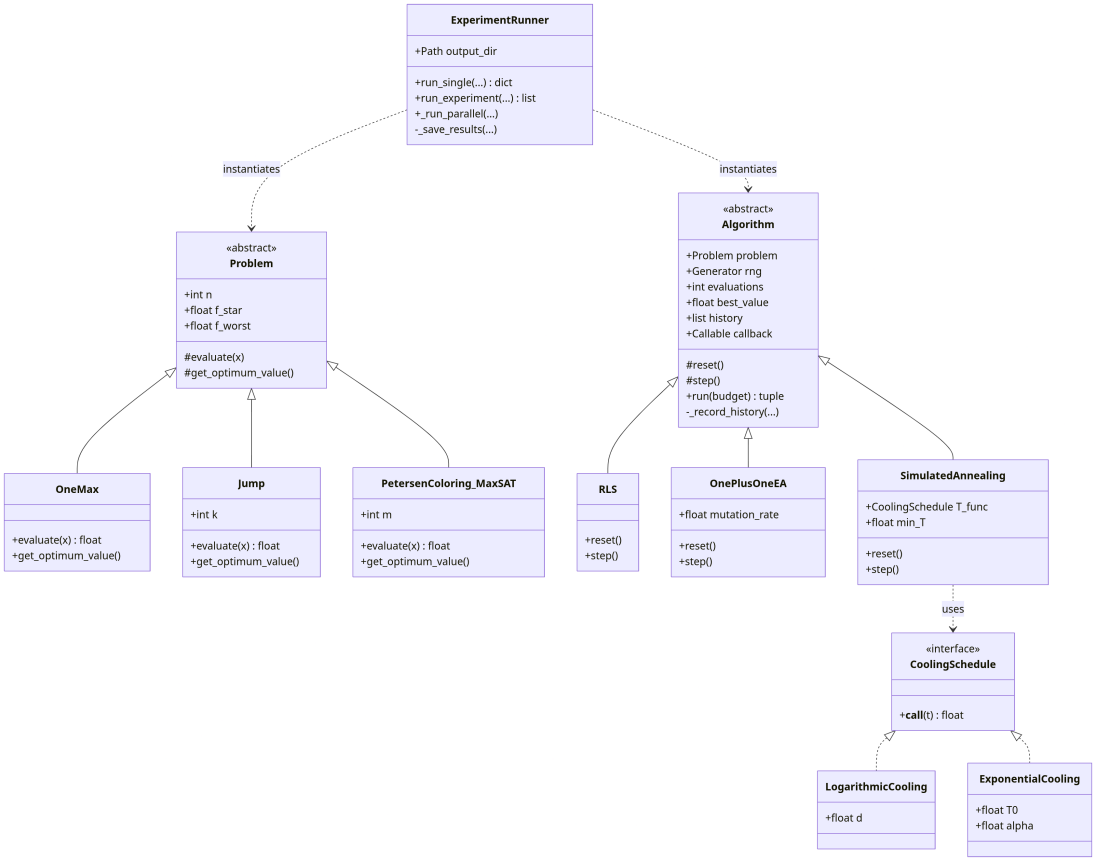
```
results/raw/<suite>/<problem>/<algorithm>/n<N>/b<B>.json
```

where `<suite>`, `<problem>`, and `<algorithm>` are normalized display names, `N` is the problem dimension, and `B` is the evaluation budget. Each file contains three top-level keys: `metadata` (algorithm, problem class, dimension, budget, run count, mode, and timestamp), `statistics` (mean, median, standard deviation, quartiles, and probability of optimum across all runs), and `results` (an ordered list of per-run dictionaries each containing regret, best value, optimality flag, evaluation count, seed, and the complete trajectory (in `full` mode)).

Generated figures are written under `results/figures/` following an analogous hierarchy:

```
results/figures/<suite>/<problem>/n<N>/<subdir>/<filename>.pdf
```

Figure 3.2: Simplified UML class diagram of the `regret` framework. Solid triangle arrows (▷) denote inheritance; dashed arrows denote dependency or instantiation. Hashes '#' indicate abstract declarations. Several concrete subclasses (e.g. `MuPlusLambdaEA`, `RLSExploration`, `NKLandscape`, `LinearCooling`) are omitted for clarity.



Plot subdirectories are configurable via `layout.structure` in the YAML configuration; the defaults are `aggregate/` for cross-budget plots, `history/` for trajectory and regret time-series plots, and `distributions/` for budget-specific box-plots and performance profiles. Runtime profile plots are written to a `profiles/` subdirectory within the raw results tree. All figures are saved as PDF at 300 DPI. Table 3.2 summarizes the full output organization.

Table 3.2: Output file organization for a representative experiment. "Normalization" replaces spaces, parentheses, and special characters with underscores or removes them.

| **Path pattern** | **Contents** |
|---|---|
| *Raw data (JSON)* | |
| `results/raw/<suite>/<prob>/<algo>/n<N>/b<B>.json` | Per-budget result file: metadata, aggregate statistics, per-run regrets, and (in `full` mode) full evaluation trajectories. |
| `results/raw/<suite>/<prob>/profiles/n<N>/cr_profile_data_<algo>.csv` | Runtime profile CSV; columns: `evaluation`, `expected_cumulative_regret`, `mean_cumulative_regret`. |
| *Figures (PDF, 300 DPI)* | |
| `.../n<N>/aggregate/` | Cross-budget plots: regret curves, convergence probability, comparison heatmap. |
| `.../n<N>/history/` | Trajectory plots: best/current value, instantaneous and cumulative regret, Time to First Optimum (TTFO) distribution. |
| `.../n<N>/distributions/` | Budget-specific plots: regret box-plots and performance profiles. |
| `.../n<N>/profiles/` | Runtime profile plots: inverse profile surface/curves, $\mathbb{E}[\mathrm{CR}(T)]$ verification. Requires `suite.profile: true`. |

Table 3.3: Plot types generated across experiments. "Profile required" denotes plots that require `suite.profile: true`

| **Plot** | **Output filename pattern - `pdf` format** | **Profile required** |
|---|---|---|
| Regret curves | `regret_curves` | No |
| Convergence probability | `convergence_probability` | No |
| Comparison heatmap | `comparison_heatmap` | No |
| Regret box-plots | `regret_boxplot_b{budget}` | No |
| Performance profile | `performance_profile_b{budget}` | No |
| History (best-so-far) | `history_best` | No |
| Instantaneous regret | `history_regret_instantaneous` | No |
| Cumulative regret | `history_regret_cumulative` | No |
| TTFO distribution | `history_ttfo_markers` | No |
| Runtime profile surface | `inverse_runtime_profile_surface_{alg}` | Yes |
| Runtime profile curves | `inverse_runtime_profile_curves` | Yes |
| CR profile verification | `cr_profile_verification` | Yes |

## 3.2 Testing strategy

The test suite follows a Test-Driven Development (TDD) discipline using a red–green–refactor workflow: tests were written prior to or alongside implementation and serve as executable specifications [1, 12]. Coverage is measured using the `coverage` [7] package with branch tracking enabled, and a minimum threshold of **50% branch coverage** is enforced in Continuous Integration (CI) via GitHub Actions. Coverage is also computed across Python 3.12, 3.13, and 3.14 in Continuous Integration (CI) for redundancy and support (the project must

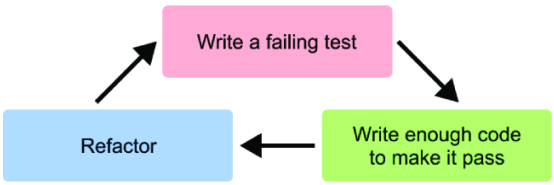



Figure 3.3: Red–Green–Refactor TDD technique. Start by writing a failing test (the red step). Next, write code that is just enough to make the written test pass (the green step). Finally, as the regularities and properties of the problem space are understood, refactors can be implemented that keeps the code clean (the blue step). The refactor step is safe since it will always be backed up by tests. [12]

be compatible for `python` $\geq$ 3.12).

Test organization follows a selective weighting strategy: core algorithmic components (`test_algorithms.py`, `test_metrics.py`, `test_problems.py`) and integration paths (`test_integration.py`) receive dense coverage, while infrastructure components such as CLI wiring and plot dispatch are covered at a coarser granularity. Shared fixtures in `conftest.py` provide canonical problem instances and bitstrings used across test suites.

## 3.3 Project Dependencies

The project is built with `hatchling` [27] and managed with `uv` [3], a Rust-based package and environment manager that replaces the Python standard `pip`/`venv`. Runtime installation: `uv sync`; development installation: `uv sync --group dev`.

Table 3.4: Package dependencies for the `regret` framework (Python $\geq$ 3.12).

| **Package** | **Min. version** | **Purpose** |
|---|---|---|
| *Scientific computing* | | |
| `numpy` [26] | 2.4.3 | Array operations, random number generation |
| `scipy` [53] | 1.16.3 | Statistics, drift computations, profile integrals |
| `pandas` [45] | 2.3.3 | Results aggregation, tabulation, CSV export |
| *Visualisation* | | |
| `matplotlib` [32] | 3.10.8 | All static plot generation |
| `ipympl` | 0.10.0 | Interactive Matplotlib in notebooks |
| `networkx` [25] | 3.6.1 | Graph problem instances (e.g. Petersen colouring) |
| *Configuration and CLI* | | |
| `pyyaml` [51] | 6.0.3 | Experiment configuration parsing |
| `jsonschema` [46] | 4.26.0 | YAML schema validation |
| `typer` [48] | 0.24.1 | `run_experiment` CLI |
| `tqdm` [15] | 4.67.1 | Progress bars for parallel runs |
| *Notebook and interactive demonstration* | | |
| `jupyter` [24] | 1.1.1 | Notebook server |
| `ipykernel` | 7.2.0 | Jupyter kernel support |
| `ipywidgets` | 8.1.8 | Notebook UI widgets |
| *Development — linting (optional)* | | |
| `ruff` [2] | — | Fast Rust-based linter and formatter (E, W, F, I, B, UP rules) |
| *Development — testing (optional)* | | |
| `pytest` [38] | — | Unit and integration test runner |
| `coverage` [7] | — | Branch coverage reporting ($\geq$50% enforced) |
| `tabulate` | — | Tabular output in test fixtures |
| *Development — type checking (optional)* | | |
| `ty` [4] | — | Rust-based static type checker (Astral) |
| *Development — documentation (optional)* | | |
| `zensical` | — | Documentation site generator |
| `mkdocstrings-python` | | Auto-generated API reference from docstrings |

# Chapter 4

# Runtime Profiles and Anytime Performance

## 4.1 Motivation

Section 2.2.6 surveyed the three standard frameworks for measuring the performance of black-box optimization algorithms: the expected runtime $\mathbb{E}[\tau_{f^*}]$, fixed-budget analysis, and fixed-target analysis culminating in the runtime profile framework from [47]. Each improves on its predecessor in descriptive richness, yet none provides a single scalar that captures solution quality at every point in the run. Fixed-budget analysis, for instance, compresses the full trajectory to one value per budget [35]. What the practitioner actually observes is the evolution of solution quality over the *entire* course of the run.

Pinto and Doerr make the case for this richer mode of reporting directly:

> Our intention is rather to highlight to researchers working on theoretical aspects in evolutionary computation that, beyond being possibly more relevant for practitioners, explicitly reporting runtime profiles can give quite interesting insights into the performance of EAs. ([47], p. 19)

This chapter takes that framing as its starting point. Building on the notation established in Section 2.2.6.3, this chapter formalizes the distributional runtime profile as a mathematical object, and then proves that expected cumulative regret decomposes exactly as a double sum over this profile. The result unifies the runtime-analysis and regret-based perspectives into a single identity, providing

the *combined anytime measure* Pinto and Doerr [47] call for.

## 4.2 Variants

### 4.2.1 Expectation-Based Runtime Profiles

Building on the first hitting time $\tau_v$ defined in Section 2.2.6.3, Pinto and Doerr [47] propose reporting the full collection of expected first hitting times $\big(\mathbb{E}[\tau_v]\big)_v$ across all relevant target levels, the *expectation-based runtime profile*. When canonical fitness levels exist, such as for OneMax or LeadingOnes, these provide natural targets; for other functions, intermediate values identified during analysis or by linear interpolation of the fitness range may be used [47]. No new algorithms or proof techniques are required: the profile is typically already implicit in existing runtime proofs and calls only for a richer mode of presentation.

The practical value of this richer view is illustrated by the behavior of $(1+1)$ EA variants on LeadingOnes: reporting only the expected optimization time suggests RLS is superior, yet the runtime profile reveals that $(1+1)$ $\mathrm{EA}_{>0}$ (*App. para.* A.2) reaches all intermediate fitness levels $i \leq 7{,}980$ faster than RLS for problem size $n = 10{,}000$, with RLS only pulling ahead in the final portion of the run [47]. Such crossover behavior is directly relevant to the design of adaptive parameter and operator selection schemes.

### 4.2.2 Distributional Runtime Profiles

The expectation-based runtime profile of [47] summarizes the distribution of $\tau_v$ by its mean. A finer characterization, and the appropriate quantity for connecting runtime profiles to regret-based measures, is the *distributional runtime profile*

$$\mathcal{P}(v,t) \;:=\; \mathbb{P}(\tau_v > t), \tag{4.1}$$

which records, for each target level $v$ and time step $t$, the probability that level $v$ has not yet been reached by time $t$. The expectation-based profile is recovered by the standard tail-sum formula for non-negative integer-valued random variables [16, p. 16]:

$$\mathbb{E}[\tau_v] \;=\; \sum_{t=0}^{\infty} \mathbb{P}(\tau_v > t). \tag{4.2}$$

The distributional profile is the more primitive object from which expectation-based statements follow, and it constitutes the formal link between runtime profiles and the cumulative regret framework developed below.

**Remark 2** (Naming convention)**.** *The term* runtime profile *appears in two distinct senses in the literature and in this work. [47] uses it to denote the* expectation-based *runtime profile* $\left(\mathbb{E}[\tau_v]\right)_v$*, which summarizes per-level performance by expected hitting times. The present work additionally employs the strictly finer* distributional *runtime profile* $\mathcal{P}(v,t) = \mathbb{P}(\tau_v > t)$*, from which the expectation-based profile is recovered via the tail-sum formula (Equation 4.2). To avoid ambiguity, both notions are referred to explicitly whenever both are in scope. Hereafter, and in all practical experimentation in this report,* runtime profile *refers to the distributional variant* $\mathcal{P}(v,t)$ *unless otherwise stated.*

## 4.3 Connection to Cumulative Regret

One of the directions suggested by Pinto and Doerr in their *Future Works* section is restated for emphasis:

> We feel that there is a need for combined measures that are capable of describing the anytime behavior of an EA. Regret-based measure as used in machine learning could be a key here, but we haven't been able so far to identify a fully satisfying measure. ([47], p. 22)

As far as we know, establishing such a relationship remains an open question. This section consolidates the following proposal: that the *Cumulative Regret (CR)* (Section 2.4.2.2) and the distributional runtime profile are not independent approaches to algorithm performance but are two representations of the same underlying quantity. The identity below attempts to establish this relation.

The relationship can be made explicit through a formal setup. A randomized black-box algorithm produces an adaptive sequence of query points $x_1, x_2, \ldots \in \{0,1\}^n$. Recall from Section 2.4.2.2 the *best-so-far process*

$$B(t) := \max_{s \le t} f(x_s),$$

which records the best objective value seen up to time $t$. Since $B(t)$ is non-decreasing in $t$, the event $\{B(t) < v\}$ is equivalent to $\{\tau_v > t\}$: the level $v$ has not

yet been hit. The incumbent cumulative regret over a budget of $T$ evaluations is therefore

$$\mathrm{CR}(T) := \sum_{t=1}^{T} \Big( f^* - B(t) \Big).$$

The proof of the main theorem relies on two elementary lemmas.

**Lemma 1** (Tail-sum representation for the gap)**.** *For any integer $b$ with $0 \le b \le f^*$,*

$$f^* - b = \sum_{v=1}^{f^*} \mathbf{1}[b < v].$$

*Proof.* The right-hand side counts the integers $v \in \{1, \dots, f^*\}$ that strictly exceed $b$, namely the set $\{b+1, \dots, f^*\}$, which has cardinality $f^* - b$. □

**Lemma 2** (Indicator equivalence)**.** *For all $v \in \{1, \dots, f^*\}$ and $t \in \mathbb{N}$,*

$$\mathbf{1}[B(t) < v] = \mathbf{1}[\tau_v > t].$$

*Proof.* Since $B(t)$ is non-decreasing in $t$, the event $\{B(t) \ge v\}$ is equivalent to $\{\tau_v \le t\}$. Taking complements gives $\{B(t) < v\} = \{\tau_v > t\}$, so the two indicators agree on every sample path. □

The following identity connects *expected* cumulative regret to the distributional runtime profile.

**Theorem 3** (Tail-Sum Regret Representation)**.** *For any randomized black-box algorithm operating on $f$ with finite budget $T \in \mathbb{N}$,*

$$\mathbb{E}[\mathrm{CR}(T)] = \sum_{v=1}^{f^*} \sum_{t=1}^{T} \mathbb{P}(\tau_v > t).$$

*Proof.* For each $t$, Lemma 1 applied to $b = B(t)$ gives

$$f^* - B(t) = \sum_{v=1}^{f^*} \mathbf{1}[B(t) < v].$$

Summing over $t = 1, \dots, T$ yields

$$\mathrm{CR}(T) = \sum_{t=1}^{T} \Big( f^* - B(t) \Big) = \sum_{t=1}^{T} \sum_{v=1}^{f^*} \mathbf{1}[B(t) < v].$$

By Lemma 2, $\mathbf{1}[B(t) < v] = \mathbf{1}[\tau_v > t]$ on every sample path, so

$$\mathrm{CR}(T) \;=\; \sum_{t=1}^{T} \sum_{v=1}^{f^*} \mathbf{1}[\tau_v > t].$$

Taking expectations and applying linearity of expectation together with $\mathbb{E}[\mathbf{1}[\tau_v > t]] = \mathbb{P}(\tau_v > t)$ gives

$$\mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{t=1}^{T} \sum_{v=1}^{f^*} \mathbb{P}(\tau_v > t).$$

Since both sums are finite, their order may be exchanged, yielding the stated result. □

Theorem 3 expresses expected cumulative regret as a double sum of the distributional runtime profile: $\mathbb{E}[\mathrm{CR}(T)]$ is the total expected time the algorithm spends below each fitness level $v$, summed over all levels. An algorithm with a uniformly smaller runtime profile (one that reaches each fitness level earlier in expectation) therefore incurs lower cumulative regret. The regret measure thus simultaneously rewards early progress at every tier of the landscape, formalizing the intuition that steady improvement throughout a run is preferable to stagnation followed by a late breakthrough, even when both strategies share the same expected optimization time.

Taking the infinite-horizon limit, and applying the tail-sum identity for $\mathbb{E}[\tau_v]$ along with the monotone convergence theorem, yields the following asymptotic connection.

**Corollary 4** (Asymptotic Regret and Expected Hitting Times)**.** *Suppose that $\tau_v$ is almost surely finite with finite expectation for each $v \in \{1, \ldots, f^*\}$. Then*

$$\lim_{T\to\infty} \mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \mathbb{E}[\tau_v] \;-\; f^*.$$

*Proof.* **Step 1: Tail-sum identity for $\mathbb{E}[\tau_v]$.** Since $\tau_v \in \mathbb{N}_{\geq 1}$ by the convention that $t = 1$ is the first oracle call and initialization is excluded from the query budget, we have $\mathbb{P}(\tau_v > 0) = 1$. The standard tail-sum formula for a positive integer-valued random variable then gives

$$\mathbb{E}[\tau_v] \;=\; \sum_{t=0}^{\infty} \mathbb{P}(\tau_v > t) \;=\; 1 + \sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t),$$

hence $\sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t) = \mathbb{E}[\tau_v] - 1$.

**Step 2: Passing to the limit in Theorem 3.** For each fixed $v$, the partial sums $\sum_{t=1}^{T} \mathbb{P}(\tau_v > t)$ are non-decreasing in $T$ and non-negative, so the monotone convergence theorem gives

$$\lim_{T\to\infty} \sum_{t=1}^{T} \mathbb{P}(\tau_v > t) \;=\; \sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t) \;=\; \mathbb{E}[\tau_v] - 1.$$

**Step 3: Summing over $v$.** Applying Step 2 to each $v = 1, \ldots, f^*$ and invoking Theorem 3,

$$\lim_{T\to\infty} \mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \Big(\mathbb{E}[\tau_v] - 1\Big) \;=\; \sum_{v=1}^{f^*} \mathbb{E}[\tau_v] \;-\; f^*. \qquad \square$$

Corollary 4 provides a unifying perspective: the long-run expected cumulative regret decomposes as the sum of the expected first hitting times over all fitness levels, minus the constant $f^*$ arising from the $t = 0$ term in the tail-sum identity. Algorithms that reach all intermediate fitness levels quickly are therefore those that incur low long-run cumulative regret. This formalizes the intuition advanced in [47] that an algorithm making steady progress through the fitness landscape is preferable to one that stagnates at low fitness values even if both ultimately find the optimum in the same expected time.

The identity also reveals that the regret-based measures studied in the online learning literature [13] and the runtime profile framework of [47] are not independent approaches to measuring algorithm performance but are two views of the same underlying quantity, connected through the distributional runtime profile $\mathcal{P}(v, t)$.

## 4.4 Practical Relevance

As discussed in Section 2.2.6.4, the limitations of classical performance measures stem from their treatment of optimization as a sequence of threshold events rather than as a continuous quality trajectory.

Pinto and Doerr [47] note that runtime profiles are particularly valuable for the design of non-static parameter and operator selection strategies in EAs (such as adaptive mutation rates, crossover probabilities, and selection pressure) since such strategies must make decisions based on current solution quality rather than

on the expected optimization time alone. The identification of *crossover fitness levels* (where one algorithm overtakes another in the runtime profile) provides actionable guidance for when to switch between strategies.

Furthermore, as established in Theorem 3, any collection of upper bounds on $\mathbb{E}[\tau_v]$ for all levels $v$ simultaneously yields an upper bound on expected cumulative regret, and conversely, lower bounds on regret propagate to lower bounds on a weighted aggregate of first hitting times. Cumulative regret thus provides a single scalar summary of the entire runtime profile, complementing the per-level view that the profile itself offers. The two representations (regret as a scalar, runtime profile as a function) are therefore best understood as complementary instruments for the same measurement task, connected by the identity of Theorem 3.

# Chapter 5

# Experiments

## 5.1 Experiment Structure

All experiments share a common infrastructure defined in the YAML configuration files under `configs/experiments/` !TODO: Cite appendices!. Each configuration specifies four top-level blocks: `suite` (global settings), `problems` (benchmark functions), `algorithms` (optimizers), and `plotting` (output).

### 5.1.1 Repetitions and budget

Each algorithm–problem pair is run for $R$ independent repetitions (typically 25–100) at budgets drawn from a per-suite budget list. The display horizon `budget_for_plots` is chosen to be roughly twice the expected time-to-first-optimum of the hardest algorithm on each problem, so trajectory plots show both approach and post-convergence behavior.

### 5.1.2 Execution mode

All suites reported in this chapter were run in `full` mode with parallelism enabled. Full mode records complete per-evaluation trajectories, which are required for the instantaneous-regret, cumulative-regret, and TTFO analyses, and for all runtime-profile computations. Although the framework also supports lighter execution modes, those are outside the scope of the present experiments because they do not preserve the trajectory-level data needed for the chapter's main measurements.

### 5.1.3 Visualization and Table Strategy

Each suite produces plots and summary tables from the stored JSON results. Table 5.1 lists every plot type with its key, diagnostic purpose, and the suites or conditions that enable it.

| Group | Output key | Diagnostic purpose | $y$-axis |
|---|---|---|---|
| Aggregate | `regret_curves` | Mean simple regret vs. budget; crossover budget between algorithms | log |
| | `convergence_probability` | $P(\text{opt})$ vs. budget; separates partial from full convergence | linear |
| | `comparison_heatmap` | $\log_{10}(\text{mean regret})$ colour-coded over all alg–budget cells | — |
| Distribution | `regret_boxplots` | Run-level spread of simple regret; skewness from non-converging runs | linear |
| | `performance_profile` | Empirical CDF of regret on log axis; steep rise = reliable convergence | log |
| Trajectory | `history_best` | Monotone incumbent fitness with $f^*$ reference | linear |
| | `regret_instantaneous` | IR of evaluated solution; fluctuates when worsening moves accepted | log |
| | `regret_instantaneous_best` | IR of incumbent (monotone non-increasing); non-zero floor = saturation | log |
| | `regret_cumulative` | Integral of current-solution IR; penalises slow convergence | linear |
| | `regret_cumulative_best` | Integral of incumbent IR; key signal in barrier-crossing suites | linear |

| | | | |
|---|---|---|---|
| | `ttfo_distribution` | Jittered TTFO samples per algorithm; conditioned on convergence | linear |
| Runtime profile (suites 03, 08, 12) | `inverse_runtime_profile_surface` | $P(\tau_v \leq T)$ heat map over $(T, v)$; saturation frontier visible | — |
| | `inverse_runtime_profile_curves` | $P(\tau_v \leq T)$ vs. $T$ at $f^* \times \{0.25, 0.5, 0.75, 0.95\}$ | linear |
| | `cr_profile_verification` | Overlay of direct and profile-derived $\mathbb{E}[\mathrm{CR}(T)]$ | linear |
| Summary | per-problem table | Mean, median, SD, IQR, 95% CI, $P(\mathrm{opt})$, run count per algorithm | — |
| | aggregate table | Across-problem mean regret, mean $P(\mathrm{opt})$, best-count per algorithm | — |

Table 5.1: Plot types produced by the experimental pipeline. The "Output key" column matches the YAML `plots` sub-block name. All trajectory and distribution plots are generated at `budget_for_plots`; aggregate plots pool across all budgets in the suite list. IR = instantaneous regret; CR = cumulative regret.

For hill-climbing algorithms (RLS, $(1+1)$-EA) the current-solution and incumbent IR variants coincide because no worsening moves are accepted. For SA and RLSExploration the two diverge during active exploration, making the current-solution variant a direct diagnostic of temperature-driven behaviour. Cumulative regret is always plotted on a linear $y$-axis: its slope during the trapped phase grows linearly with time, and a log axis would suppress that signal.

### 5.1.4 Suite Numbering and Roles

Suites are numbered 01–14. Table 5.2 gives a complete reference; the methodological context for runtime-profile suites (03, 08, 12) is developed in Chapter 4.

| Suite | Role | Problems | Algorithms | $R$ | Max budget | Profile |
|---|---|---|---|---|---|---|
| 01 | Baseline comparison | 6 pseudo-boolean | RLS, RLSExpl, EA, SA-Log-d2.0 | 50 | 10 000 | No |
| 02 | Extended benchmark | 9 pseudo-boolean + combinatorial | same 4 families (with overrides) | 50 | 20 000 | No |
| 08 | Tail-sum verification | OneMax, LeadingOnes | RLS, $(1+1)$-EA | 100 | 10 000 | **Yes** |
| 03 | NK ruggedness ($6 \times 5$ grid) | 5 NK ($k \in \{1,2,4,6,8\}$, $n = 20$) | RLS, EA, 6×SA-Log | 25 | 50 000 | **Yes** |
| 04 | Trap: deception width × rate | 4 Trap ($k \in \{5,8,10,20\}$) | 4×EA + SA-Log-d4.0 | 25 | 10 000 | No |
| 05 | Jump: mutation rate / plateau onset | 5 Jump ($k \in \{2,3,5,7,10\}$) | 3×EA (fixed + adaptive $k/n$) | 25 | 20 000 | No |
| 06 | Plateau: neutral walk × rate | 4 Plateau ($k \in \{2,5,10,20\}$) | 3×EA + SA-Log-d4.0 | 30 | 20 000 | No |
| 07 | HIFF scaling | 3 HIFF ($n \in \{32,64,128\}$) | RLS, 4×EA, SA-Log-d4.0 | 25 | 50 000 | No |
| 09 | Jump: plateau duration (extended) | 5 Jump (same as 05) | 2×EA (fixed + adaptive) | 25 | 50 000 | No |
| 10 | Barrier archetype comparison | Jump/Trap/Plateau ($k = 5$, $n = 100$) | 4×EA ($0.5/n$–$5/n$) | 30 | 50 000 | No |
| 11 | SA cooling diagnostics | OneMax, LeadingOnes | RLS, 6×SA-Log, SA-Lin, SA-Exp | 50 | 10 000 | No |
| 12 | NK saturation | 5 NK ($k \in \{0,2,4,6,8\}$, $n = 20$) | RLS, $(1+1)$-EA, SA-Log-d4.0 | 25 | 50 000 | **Yes** |
| 13 | Regret scaling with $n$ | 4 OneMax ($n \in \{25,50,100,200\}$) | $(1+1)$-EA at rate $1/n$ | 60 | 10 000 | No |
| 14 | Population dynamics | OneMax, LeadingOnes, TwoMax, Jump-3 | $(1+1)$-EA + 4×$(\mu+\lambda)$-EA | 30 | 10 000 | No |

Table 5.2: Overview of all 14 experimental suites. $R$ = runs per algorithm–problem–budget combination. "Profile" indicates whether `suite.profile: true` is set. Budget values are in evaluations; "max" is the largest value in the suite budget list.

## 5.2 Baseline Suites

### 5.2.1 Suite 01: Algorithm-Family Comparison

Suite 01 establishes a reference comparison of four algorithm families across six pseudo-boolean problems of increasing structural difficulty. The four families are: Randomized Local Search (RLS), RLSExploration, the $(1+1)$-EA, and SA with logarithmic cooling at $d = 2.0$ (SA-Log-d_2.0). SA is reduced to a single representative schedule to preserve a clean four-family distinction; the effect of varying the schedule parameter is deferred to suites 03 and 11. All algorithm entries carry empty `by_problem` blocks, so the same configuration applies across all six problems.

The six problems are: OneMax ($n = 100$), LeadingOnes ($n = 100$), TwoMax ($n = 100$), Jump-3 ($n = 100$, $k = 3$), Trap-10 ($n = 100$, $k = 10$), and BinVal ($n = 50$). BinVal uses $n = 50$ rather than $n = 100$: its exponential fitness structure ($f^* = 2^{50} - 1 \approx 10^{15}$) already produces large fitness gaps and rapid convergence at this size, and extending to $n = 100$ would make the optimum value $2^{100} - 1$ while adding no structural insight. The much smaller `budget_for_plots` value of 500 for BinVal, compared with 2000–10 000 for the other problems, reflects the expectation that convergence occurs early.

The 50 runs per combination are evaluated at six budgets: $\{200, 500, 1000, 2000, 5000, 10000\}$. The upper budget of 10 000 is chosen to include the regime where LeadingOnes and Jump-3 actually converge for some algorithms; the lower bound of 200 captures the very-early transient. Pairwise statistical annotations are enabled on regret box-plots, performance profiles, and TTFO distributions, using RLS as the reference algorithm.

### 5.2.2 Suite 02: Extended Pseudo-Boolean Benchmark

Suite 02 extends the baseline to nine problems, adding Jump-5, Trap-5, Plateau-10, HIFF-64, and a combinatorial instance (Petersen-3-Color). The algorithm set is identical to suite 01 (the same four families). Budgets range from 500 to 20 000 over 50 runs. Profile analysis is disabled.

Table 5.3 details the problem-specific hyperparameter overrides applied in this suite. The override problems — Jump-3, Jump-5, and Trap-5 — share the structural property of having barriers that require sustained exploration; the combinatorial instance and the unimodal and bimodal problems receive no overrides.

For RLSExploration, fixing $\varepsilon = 0.05$ and disabling decay holds the exploration rate constant at five times the default, maintaining persistent non-improving moves on problems where barrier crossing requires sustained exploration. For SA, tripling $d$ under

Table 5.3: Problem-specific algorithm overrides in suite 02. For RLSExploration the default is $\varepsilon = 1/n = 0.01$ with exponential decay ($\varepsilon_t = \varepsilon/t$); for SA the default cooling parameter is $d = 2.0$.

| **Algorithm** | **Problem** | **Parameter** | **Default** | **Override** |
|---|---|---|---|---|
| RLSExploration | Jump-3 | $\varepsilon$, decay | 0.01, on | 0.05, off |
| | Jump-5 | $\varepsilon$, decay | 0.01, on | 0.05, off |
| | Trap-5 | $\varepsilon$, decay | 0.01, on | 0.05, off |
| SA-Log-d_2.0 | Jump-5 | $d$ | 2.0 | 6.0 |
| | Trap-5 | $d$ | 2.0 | 6.0 |

logarithmic cooling $T(t) = d/\log(t+1)$ triples the temperature at every evaluation index, raising acceptance probability for worsening moves throughout the run. Both modifications target the same function: improving escape from local or deceptive optima at moderate budget.

## 5.3 Verification Suite

### 5.3.1 Suite 08: Runtime Profile Verification

Suite 08 validates the tail-sum identity

$$\mathbb{E}[\mathrm{CR}(T)] = \sum_{v=1}^{f^*} \sum_{t'=1}^{T} [1 - P(\tau_v \leq t')]$$

on two problems with well-understood convergence behavior: OneMax and LeadingOnes ($n = 100$), using only RLS and the $(1+1)$-EA.

The suite is configured with 100 independent runs, more than any other suite. This is not for statistical power in the usual sense (the goal is not to detect a treatment effect) but to suppress Monte-Carlo variance in the empirical mean $\widehat{\mathbb{E}}[\mathrm{CR}(T)]$ to the point where any visible gap between the trajectory-derived and profile-derived curves in the verification plot can be attributed to numerical discretization rather than sampling noise. With $R = 100$ runs and the integer-grid profile implementation (exact for unit-increment fitness), any remaining discrepancy indicates a bug in trajectory recording, cumulative-regret computation, or profile construction.

Profile analysis is enabled (`profile: true`). The `cr_profile_verification` plot is the primary output: it overlays the two derivations of $\mathbb{E}[\mathrm{CR}(T)]$ for each algorithm on a shared time axis. Inverse runtime profile surface and curve plots are also produced. The `budget_for_plots` for LeadingOnes is set to 10 000, which is the largest available

budget; this is set deliberately to capture the full cumulative regret ramp, which extends further than on OneMax due to the $\Theta(n^2)$ expected convergence time.

Pairwise annotations are enabled on box-plots and TTFO distributions with RLS as the reference, though these are secondary outputs given the verification focus.

## 5.4 Exploratory Analysis of Regret and Runtime

The exploratory suites address four themes: barrier structure and mutation rate (suites 04–06, 09–10), NK ruggedness and SA cooling (suites 03, 11–12), hierarchical structure and scaling (suites 07, 13), and population dynamics (suite 14). Their purpose is to examine how regret and convergence behaviour change with landscape structure, cooling schedule, problem size, and population configuration.

### 5.4.1 Suite 03: NK Landscape Ruggedness

Suite 03 crosses six logarithmic SA schedules ($d \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$) with five NK instances ($k \in \{1, 2, 4, 6, 8\}$, $n = 20$, seed 42). RLS and the $(1+1)$-EA are included as zero-temperature baselines so that the effect of temperature-driven acceptance can be separated from plain hill-climbing.

All SA entries use empty `by_problem` blocks, so the suite forms a clean $6 \times 5$ grid over cooling parameter and epistasis level without per-problem tuning. The restriction to $n = 20$ is imposed by the exact-optimum computation used in the NK implementation. Profile analysis is enabled, producing inverse runtime profile surfaces, selected profile curves, and the profile verification plot. The extended budget to 50 000 is needed because high-epistasis instances may approach saturation only late in the run.

### 5.4.2 Suite 04: Trap Function Mutation Rate

Suite 04 evaluates Trap with four $(1+1)$-EA mutation rates ($0.5/n$, $1/n$, $2/n$, $5/n$) and one SA variant across four deception widths ($k \in \{5, 8, 10, 20\}$, $n = 100$). The EA rates are fixed across all problems. SA-Log-d_4.0 uses $d = 4.0$ by default, with a single override to $d = 8.0$ on Trap-k20.

Trap-k1 is excluded because it yields uniformly uninformative behaviour: all algorithms are driven to the deceptive attractor, so the resulting plots are essentially flat. The chosen EA rates are shared with suites 05 and 06 to support cross-suite comparison of regret shape across barrier types. Profile analysis is disabled.

### 5.4.3 Suite 05: Jump Function Mutation Rate

Suite 05 studies five Jump instances ($k \in \{2,3,5,7,10\}$, $n = 100$) under three $(1+1)$-EA variants: fixed rates $1/n$ and $2/n$, and a problem-adaptive rate $k/n$ implemented through per-problem overrides.

The lower rate $0.5/n$ is omitted because it offers no advantage for crossing a $k$-bit gap relative to $1/n$. Budgets range from 500 to 20 000. Instantaneous regret is plotted on a log scale, while cumulative regret is kept on a linear scale so that the trapped-phase slope remains visible. Profile analysis is disabled; the longer-horizon Jump study is deferred to suite 09.

### 5.4.4 Suite 06: Plateau Function Mutation Rate

Suite 06 mirrors suite 05 on Plateau instances ($k \in \{2,5,10,20\}$, $n = 100$) to compare cumulative regret trajectory shape across barrier types at structurally equivalent widths. The algorithm set is three $(1+1)$-EA variants at rates $1/n$, $2/n$, and $5/n$, plus SA-Log-d_4.0 with no per-problem overrides.

The adaptive $k/n$ rate from suite 05 is omitted: it was motivated by the need to maximize the probability of exactly $k$ simultaneous flips for the Jump gap, a rationale that has no equivalent for the neutral walk on Plateau, which does not require any specific number of simultaneous flips. SA-Log-d_4.0 is added instead as a third algorithm class to assess whether temperature-driven acceptance confers any advantage on a landscape where the neutral zone carries no fitness penalty. The shared values $k = 2$ and $k = 5$ support direct comparison with suite 05; $k = 10$ and $k = 20$ extend the plateau duration beyond what can converge within the 20 000-evaluation budget. Profile analysis is disabled.

### 5.4.5 Suite 07: Hierarchical IFF Scaling

Suite 07 evaluates HIFF at $n \in \{32,64,128\}$ under RLS, four $(1+1)$-EA variants, and SA-Log-d_4.0 at a fixed $d = 4.0$. RLS is included as a zero-temperature reference whose rapid saturation on large instances anchors the comparison: a single bit-flip disrupts any uniform sub-block, so RLS is expected to stagnate at the first hierarchical level for $n \geq 64$. Table 5.4 shows the exact mutation rate values used for each EA variant at each problem size.

For $n = 128$, `budget_for_plots` is set to 50 000, the largest budget in the suite. Convergence is not expected within this budget; the setting is deliberate, making the unsolved $n = 128$ trajectory an informative contrast to the partially or fully solved $n = 32$ and $n = 64$ cases on the same plot axes. Profile analysis is disabled.

Table 5.4: Per-problem mutation rates in suite 07. Values are set via `by_problem` overrides to exact decimal representations of $\{0.5/n, 1/n, 2/n, 3/n\}$ at each problem size.

| **Algorithm** | $n = 32$ | $n = 64$ | $n = 128$ |
|---|---|---|---|
| EA-mu_0.5-by-n | $1/64 \approx 0.01563$ | $1/128 \approx 0.00781$ | $1/256 \approx 0.00391$ |
| EA-mu_1-by-n | $1/32 \approx 0.03125$ | $1/64 \approx 0.01563$ | $1/128 \approx 0.00781$ |
| EA-mu_2-by-n | $2/32 \approx 0.06250$ | $2/64 \approx 0.03125$ | $2/128 \approx 0.01563$ |
| EA-mu_3-by-n | $3/32 \approx 0.09375$ | $3/64 \approx 0.04688$ | $3/128 \approx 0.02344$ |

### 5.4.6 Suite 09: Jump Plateau Duration

Suite 09 uses the same five Jump instances and the same two EA variants as suite 05 (rates $1/n$ and $k/n$ with identical per-problem overrides), but differs in its budget configuration. The budget list is $\{1000, 5000, 10000, 25000, 50000\}$, dropping the 500-evaluation point from suite 05 and extending the maximum to 50 000. The `budget_for_plots` values are also extended: Jump-k2 uses 10 000, Jump-k3 uses 25 000, and Jump-k5, Jump-k7, Jump-k10 each use 50 000. In suite 05, Jump-k5 through Jump-k10 used 20 000.

The difference between the two suites is therefore one of time horizon rather than algorithm or problem design. Suite 05 captures plateau onset (the transition from hill-climbing to the trapped phase) for the smaller $k$ values; suite 09 extends the horizon to capture the plateau duration and, for small $k$ at the adaptive rate, potentially the plateau exit as well.

Profile analysis is disabled.

### 5.4.7 Suite 10: Barrier Structure Comparison

Suite 10 compares Jump-k5, Trap-k5, and Plateau-k5 at $n = 100$ under four $(1+1)$-EA mutation rates: $0.5/n$, $1/n$, $2/n$, and $5/n$. The same rates are used on all three barrier types, and the budget extends to 50 000.

The suite is designed to compare how regret accumulates under barriers of equal nominal width but different structure. TTFO alone is often too coarse in this regime, whereas cumulative regret can distinguish early deception, delayed barrier crossing, and neutral trapping. Pairwise annotations use `(1+1)-EA-mu_1-by-n` as the reference algorithm. Profile analysis is disabled.

### 5.4.8 Suite 11: SA Cooling Schedule Diagnostics

Suite 11 evaluates eight SA variants and one RLS baseline on OneMax and LeadingOnes ($n = 100$). The SA set consists of six logarithmic schedules ($d \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$), one linear schedule (SA-Lin), and one exponential schedule (SA-Exp), all applied uniformly across both problems with no per-problem overrides.

The linear and exponential variants use implementation defaults: linear cooling with $T_0 = 1.0$, $T_f = 0.01$, max_iter $= 10\,000$; exponential cooling with $T_0 = 1.0$, $\alpha = 0.95$. Both schedules reach their minimum temperature within the available budget (linear exactly at $t = 10\,000$, exponential effectively by $t \approx 90$ since $0.95^{90} \approx 0.01$) after which SA reduces to a near-deterministic hill-climber. Logarithmic schedules, by contrast, remain strictly positive for all finite $t$. This structural distinction is the primary hypothesis the suite is designed to expose as a signature in the instantaneous regret trajectory. Pairwise annotations use RLS as the reference so that any SA configuration underperforming plain hill-climbing is directly identifiable in the annotated plots. Profile analysis is disabled.

### 5.4.9 Suite 12: NK Saturation

Suite 12 examines five NK landscape instances ($k \in \{0, 2, 4, 6, 8\}$, $n = 20$, seed 42) under three algorithms: RLS, the $(1+1)$-EA, and SA-Log-d_4.0. The NK instances use the same fixed seed as suite 03, making the landscape realisations identical across both suites for the shared $k$ values.

Suite 12 differs from suite 03 in two structural ways. First, it includes NK-k0 ($k = 0$, fully separable), absent from suite 03. At $k = 0$ each locus contributes independently to fitness, providing a baseline against which saturation at higher $k$ can be measured. Second, the SA cooling parameter escalates with $k$ via `by_problem` overrides (Table 5.5), reflecting that deeper local optima at higher epistasis require higher temperature to maintain non-trivial acceptance probability. This tuning means suite 12 does not form a clean grid design for studying the SA cooling–epistasis interaction (that role belongs to suite 03), but gives the SA variant a reasonable operating point at each $k$ to assess whether adequate temperature maintenance changes the saturation outcome.

Table 5.5: SA-Log cooling parameter $d$ per NK instance in suite 12. Values increase monotonically with epistasis $k$ to compensate for deeper local optima.

| **NK instance** | NK-k0 | NK-k2 | NK-k4 | NK-k6 | NK-k8 |
|---|---|---|---|---|---|
| $d$ | 4.0 | 4.0 | 8.0 | 8.0 | 12.0 |

Profile analysis is enabled, producing inverse runtime profile surfaces and curves.

The `cr_profile_verification` plot is not enabled in this suite: the NK landscape uses a continuous fitness function normalized to $[0,1]$, so the integer-grid profile implementation does not apply exactly.

### 5.4.10 Suite 13: Regret Scaling with Problem Size

Suite 13 runs a single algorithm, $(1+1)$-EA at rate $1/n$, on four OneMax instances with $n \in \{25, 50, 100, 200\}$. The mutation rates are set via per-problem overrides to exact values $\{0.04, 0.02, 0.01, 0.005\}$ corresponding to $1/n$ at each size; the `defaults` block carries 0.01 as a fallback. Using a single algorithm removes algorithm-to-algorithm comparisons from the design and focuses the suite entirely on the scaling behavior of the regret metric across problem sizes. With 60 runs (the highest run count after suite 08) the empirical $\mathbb{E}[\mathrm{CR}(T)]$ estimates are suitable for fitting a scaling exponent.

The `budget_for_plots` values are $\{1000, 2500, 5000, 10000\}$ for $n \in \{25, 50, 100, 200\}$ respectively. These are approximately consistent with $5n \log_2 n$ (giving roughly 580, 1411, 3322, and 7640), but set somewhat higher to ensure that the full post-convergence plateau is also visible in the trajectory plots. The budget list $\{100, 500, 1000, 2500, 5000, 10000\}$ includes a very short 100-evaluation point, reflecting the interest in early transient behavior at small $n$.

Profile analysis is disabled.

### 5.4.11 Suite 14: Population Dynamics

Suite 14 compares the standard $(1+1)$-EA against four $(\mu+\lambda)$-EA variants on four problems: OneMax, LeadingOnes, TwoMax, and Jump-3 ($n = 100$). All five algorithm entries use mutation rate $1/n = 0.01$ with no per-problem overrides, isolating the effect of population structure from mutation rate.

The five configurations are: $(1+1)$-EA (the standard single-individual baseline via its own class); $(1+1)$-EA-framework ($\mu = 1,\ \lambda = 1$ via the MuPlusLambdaEA class); $(5+5)$-EA ($\mu = 5,\ \lambda = 5$); $(10+20)$-EA ($\mu = 10,\ \lambda = 20$); and $(20+10)$-EA ($\mu = 20,\ \lambda = 10$). Including both $(1+1)$-EA and $(1+1)$-EA-framework is deliberate: the two entries use different implementation classes but the same logical algorithm, so any difference in their results would indicate a framework overhead or implementation discrepancy rather than a population effect.

The $(\mu+\lambda)$-EA implementation records one history entry per offspring evaluation rather than per generation, which was the reason for its exclusion from suite 02. In suite 14 all algorithms are either the $(1+1)$-EA or $(\mu+\lambda)$-EA variants, so the recording granularity is consistent within the suite. The four problems span unimodal (OneMax),

sequentially structured (LeadingOnes), bimodal (TwoMax), and gap-structured (Jump-3) fitness landscapes, providing a range of settings in which the cost and benefit of maintaining a population may differ.

Profile analysis is disabled.

## 5.5 Cross-Suite Design Relationships

Several structural relationships between suites affect which results can be meaningfully compared. Table 5.6 summarizes the three principal links; the paragraphs below elaborate the design rationale.

Table 5.6: Principal cross-suite design relationships.

| Relationship | Suites | Shared | Differs |
|---|---|---|---|
| Barrier types at equal rates | 04, 05, 06, 10 | EA rates $1/n$ and $2/n$; problems at same nominal $k$ | Barrier type (Trap/Jump/Plateau); additional rates per suite |
| Jump at two time horizons | 05, 09 | All 5 Jump instances; EA variants at $1/n$ and $k/n$ | Budget list and `budget_for_plots` (onset vs. duration) |
| NK at two SA designs | 03, 12 | $n = 20$, seed 42; shared $k \in \{2,4,6,8\}$; identical landscapes | Suite 03: $6 \times 5$ $d$-grid, no tuning; Suite 12: per-$k$ $d$ escalation, adds $k = 0$ |

**Barrier types across suites 04–06 and 10.** Suites 04, 05, and 06 share the fixed rates $1/n$ and $2/n$. The full rate sets differ: suite 04 uses $\{0.5/n, 1/n, 2/n, 5/n\}$; suite 05 uses $\{1/n, 2/n\}$ plus the adaptive $k/n$ (with $0.5/n$ omitted as dominated and $5/n$ absent); suite 06 uses $\{1/n, 2/n, 5/n\}$ (with $0.5/n$ absent for the same reason). At the shared rates, results on Trap-$k$, Jump-$k$, and Plateau-$k$ at the same $k$ are directly comparable since the algorithm is otherwise identical. Suite 10 consolidates this comparison by running all four EA rates on all three barrier types simultaneously at $k = 5$.

**Jump across suites 05 and 09.** Suites 05 and 09 are identical in problem and algorithm configuration; they differ only in budget list and `budget_for_plots`. Suite 05 is oriented toward observing plateau onset at small $k$; suite 09 extends the horizon

to study plateau duration at all $k$ values, including those where convergence within budget is unlikely.

**NK across suites 03 and 12.** Both suites use $n = 20$, seed 42. Suite 03 covers $k \in \{1, 2, 4, 6, 8\}$ and suite 12 covers $k \in \{0, 2, 4, 6, 8\}$; the shared values are $k \in \{2, 4, 6, 8\}$, and the landscape realisations at those values are identical. Suite 03 forms a clean grid over $d$ values with no per-problem tuning, making any cooling–epistasis interaction attributable to problem structure rather than instance-specific calibration. Suite 12 uses per-$k$ $d$ escalation and adds the $k = 0$ separable baseline, providing a different perspective on the saturation phenomenon.

# Chapter 6

# Evaluation & Discussion of Results

## 6.1 Overview

## 6.2 Baseline Suites

### 6.2.1 Suite 01: Algorithm-Family Comparison

#### 6.2.1.1 Aggregate Regret and Convergence

#### 6.2.1.2 Regret Distributions and TTFO

#### 6.2.1.3 Trajectory Analysis

### 6.2.2 Suite 02: Extended Pseudo-Boolean Benchmark

#### 6.2.2.1 Aggregate Regret and Convergence

#### 6.2.2.2 Effect of Algorithm Overrides

#### 6.2.2.3 Trajectory Analysis

## 6.3 Verification Suite

### 6.3.1 Suite 08: Runtime Profile Verification

#### 6.3.1.1 Tail-Sum Identity: Direct vs. Profile-Derived $\mathbb{E}[\mathrm{CR}(T)]$

#### 6.3.1.2 Inverse Runtime Profile Surfaces and Curves

## 6.4 Exploratory Analysis

### 6.4.1 Barrier Structure and Mutation Rate

#### 6.4.1.1 Suite 04: Trap – Deception Width and Mutation Rate

# Chapter 7

# Conclusion & Future Work

## 7.1 Conclusion

## 7.2 Future Work

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# Appendix A

# Evolutionary Algorithms, Extended

## A.1 Exact Runtime Formulae

For mutation probability $p$, the exact expected optimization time on LeadingOnes is [9]

$$\mathbb{E}[T_p] = \frac{1}{2p^2}\big((1-p)^{-n+1} - (1-p)\big),$$

minimized at $p \approx 1.59/n$, giving $\approx 0.77n^2$. For OneMax, the sharp expansion at $p = 1/n$ is [33]

$$\mathbb{E}[T] = e\,n\ln n - 1.8925\ldots n + \tfrac{e}{2}\ln n + O(1).$$

More generally, for $p = c/n$ the expected time on any linear function is $(1 \pm o(1))\frac{e^c}{c}\,n\ln n$, minimized at $c = 1$ [54].

## A.2 Implementation-Aware Variants

A practical observation of Pinto and Doerr [47] is that roughly a $1/e$ fraction of all $(1+1)$ EA iterations flip *no* bit (the offspring equals the parent), yet standard analyses charge a full function evaluation for each such iteration. Real implementations avoid these wasted evaluations by one of two strategies.

**Resampling variant – $(1+1)$ EA$_{>0}$.** The mutation strength $\ell$ is drawn from the *conditional* binomial distribution $\mathrm{Bin}_{>0}(n,p)$, i.e., $\mathrm{Bin}(n,p)$ conditioned on $\ell > 0$. This effectively skips any iteration that would flip nothing.

**Shift variant** – $(1+1)$ **EA**$_{0\to1}$**.** If $\ell=0$ is sampled, it is replaced by $\ell=1$, shifting all probability mass from zero flips to exactly one flip.

Both variants are analyzed in full in [47, 34]. The key runtime consequences with $p=1/n$ are summarized in Table A.1.

| **Algorithm** | **OneMax** | **LeadingOnes** |
|---|---|---|
| $(1+1)$ EA | $en\ln n+O(n)\approx 2.718\,n\ln n$ | $\approx 0.859\,n^2$ |
| $(1+1)$ EA$_{>0}$ | $\approx 1.718\,n\ln n+O(n)$ | $\approx 0.543\,n^2$ |
| $(1+1)$ EA$_{0\to1}$ | $\approx 1.359\,n\ln n+O(n)$ | $\approx 0.517\,n^2$ |
| RLS | $nH_n\approx n\ln n$ | $\approx 0.5\,n^2$ |

Table A.1: Expected optimization times at $p=1/n$ for the $(1+1)$ EA and its implementation-aware variants. RLS is included for reference. Bounds for the variants follow from [47, 9, 54].

Resampling or shifting yields a 35–47% reduction in function evaluations on OneMax and roughly 37–40% on LeadingOnes at $n=600$, as confirmed empirically in [47]. For the (1+1) EA$_{>0}$, the expected runtime on any linear function with $p=c/n$ becomes $(1\pm o(1))\frac{e^{c-1}}{c}\,n\ln n$ [47], a multiplicative $(1-e^{-c})$ improvement over the standard algorithm.

## A.3 General Form - '$(\mu+\lambda)$-EA'

## A.4 Genetic Algorithms

### A.4.1 Crossover

## A.5 Parameters & Operators

### A.5.1 Adaptive Mutation & crossover rates

### A.5.2 Selection pressure